Collection Risk Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Collection Risk Absorption (CRA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on CRA achieves an annualized gross (net) Sharpe ratio of 0.33 (0.26), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 2.33 (1.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor) is 23 bps/month with a t-statistic of 2.16.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This slow diffusion manifests particularly in complex accounting signals that require sophisticated analysis to interpret, where investor attention constraints and information processing frictions create temporary mispricings that resolve predictably over time.

We examine Collection Risk Absorption (CRA), a novel accounting-based signal that captures firms' exposure to collection risks embedded in their receivables and revenue recognition practices. Unlike simple financial ratios that attract immediate investor scrutiny, CRA represents a nuanced measure of operational efficiency that requires detailed analysis of working capital dynamics and customer payment patterns. This complexity suggests that information about collection risks diffuses slowly through market prices, creating a setting where initial underreaction to CRA information generates predictable return continuation as investors gradually incorporate these signals into valuations.

The slow diffusion of information provides a compelling framework for understanding why CRA predicts future returns. When firms absorb collection risks through extended payment terms or deteriorating receivables quality, this information enters the market through financial statements but faces significant barriers to immediate incorporation. Following the limited attention framework of Hirshleifer and Teoh (2003) and DellaVigna (2009), investors facing capacity constraints prioritize salient information while overlooking complex accounting signals embedded in working capital accounts. This selective attention creates systematic underreaction to CRA information, where initial price movements incompletely reflect the full implications of collection risk changes.

The gradual information diffusion mechanism operates through multiple channels that reinforce return predictability. Hong et al. (2000) demonstrate that information travels slowly across investor networks, with sophisticated investors gradually arbitraging mispricings while facing capital and attention constraints. For CRA signals, this diffusion process extends over multiple periods as analysts progressively incorporate collection risk metrics into earnings forecasts, institutional investors adjust positions based on detailed financial analysis, and market makers update pricing models to reflect receivables quality. The complexity of interpreting collection risks—requiring assessment of customer creditworthiness, industry payment cycles, and revenue recognition policies—amplifies these frictions and extends the price discovery process.

Cross-sectional variation in information frictions strengthens the predictive power of CRA where attention constraints bind most severely. Hou and Moskowitz (2007) show that return predictability concentrates among neglected stocks with limited analyst coverage and institutional ownership, where information diffusion proceeds most slowly. Similarly, Zhang (2006) documents stronger post-earnings announcement drift for firms with greater information uncertainty, suggesting that complex signals like CRA generate more pronounced underreaction when interpretation costs are high. We therefore expect CRA's return predictability to manifest most strongly among smaller firms, those with low analyst coverage, and stocks with limited institutional ownership—precisely where information processing frictions impede rapid price discovery and create exploitable return continuation patterns.

Our empirical analysis reveals that CRA strongly predicts cross-sectional returns with economically meaningful magnitudes. A value-weighted long/short trading strategy based on CRA achieves an annualized gross Sharpe ratio of 0.33, with the high-minus-low CRA portfolio generating average monthly returns of 21 basis points (t-statistic = 2.40). The strategy's performance remains robust after controlling for known risk factors, earning a monthly alpha of 22 basis points (t-statistic = 2.33) relative to the Fama-French five-factor model augmented with momentum. These results establish CRA as an economically significant predictor that captures information beyond established pricing factors.

The predictive power of CRA exhibits remarkable stability across alternative portfolio construction methodologies and persists after accounting for transaction costs. When we implement the strategy using different sorting procedures—including quintile sorts with name breaks and market capitalization breaks—the alphas consistently exceed 19 basis points per month with t-statistics above 1.94. Notably, the strategy maintains profitability net of transaction costs, generating a net Sharpe ratio of 0.26 and significant positive generalized alphas that expand the mean-variance efficient frontier for four out of five factor model specifications. Among the largest stocks (those above the 80th NYSE percentile), where arbitrage capital concentrates and information should diffuse most rapidly, the CRA strategy still achieves average returns of 25 basis points per month (t-statistic = 2.28), suggesting that collection risk information remains incompletely incorporated even in the most liquid market segments.

CRA's predictive power persists after controlling for the most closely related anomalies from the factor zoo, confirming that it captures distinct information about future returns. When we simultaneously control for six closely related predictors—including Composite debt issuance, Growth in book equity, and Asset growth—plus the Fama-French six factors, the CRA strategy maintains an alpha of 23 basis points per month (t-statistic = 2.16). Furthermore, CRA ranks in the top 30% of gross Sharpe ratios among 212 documented anomalies and improves combination strategies that aggregate multiple return predictors. These results demonstrate that CRA contains incremental information about expected returns that existing anomalies fail to capture, consistent with markets incompletely processing collection risk signals embedded in financial statements.

This paper advances the asset pricing literature by documenting a novel source of return predictability rooted in the slow diffusion of accounting information about collection risks. While prior research has identified numerous accounting-based anomalies, including the asset growth effect of Cooper et al. (2008) and the net operating assets anomaly of Hirshleifer et al. (2004), our work uniquely focuses on the granular information contained in receivables quality and collection risk absorption. Unlike broad measures of investment or accruals studied in the Journal of Financial Economics and Journal of Accounting Research, CRA captures specific operational frictions in working capital management that prior anomalies overlook. By demonstrating that collection risk information diffuses gradually through prices, we extend the limited attention literature and provide new evidence on the mechanisms generating post-announcement drift documented in Bernard and Thomas (1989).

Methodologically, we contribute to the anomaly literature by implementing the rigorous testing protocol of Novy-Marx and Velikov (2023b), ensuring that our results meet the highest standards for establishing new return predictors. Our comprehensive analysis addresses concerns about data mining and p-hacking that plague anomaly research, as highlighted in recent Journal of Finance publications examining the replication crisis. By demonstrating that CRA survives multiple robustness tests—including alternative portfolio constructions, transaction cost adjustments, and controls for 212 existing anomalies—we establish a genuinely new source of alpha that expands the investment opportunity set. The stability of CRA's performance across size groups and time periods distinguishes it from many recently proposed factors that concentrate among microcaps or fail out-of-sample tests.

Our findings carry important implications for market efficiency, accounting disclosure, and investment practice that extend beyond documenting another return predictor. The persistent underreaction to CRA information challenges semi-strong form efficiency and suggests that even sophisticated investors incompletely process complex accounting signals related to revenue quality and collection risks. For accounting standard setters and regulators featured in Journal of Accounting and Economics research, our results highlight the importance of enhancing disclosure requirements around receivables aging, customer concentration, and collection risk metrics that currently receive limited attention. From a practical perspective, the CRA anomaly's robust performance net of transaction costs and its low correlation with existing factors (ranking in the top 16% of net Sharpe ratios) indicates that it represents an implementable strategy that can improve portfolio performance, particularly when combined with other accounting-based signals in multi-factor models.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over multiple decades, allowing us to examine the evolution of collection risk absorption across different market conditions and economic cycles.Collection Risk Absorption signal utilizes two key accounting variables. Receivables estimated doubtful (RECD) represents the allowance for doubtful accounts, which is management's estimate of the portion of accounts receivable that may not be collectible. This contra-asset account reflects the firm's assessment of credit risk associated with its outstanding receivables. Nonop income other (NOPIO) captures miscellaneous non-operating income items that do not fit into other specific non-operating income categories, including various gains, recoveries, and other income sources outside the firm's core operations.construct the Collection Risk Absorption signal by computing the year-over-year change in receivables estimated doubtful scaled by lagged non-operating income other: (RECD(t) - RECD(t-1)) / NOPIO(t-1). This measure captures the relative change in a firm's provision for bad debt normalized by its non-operating income capacity. The economic intuition is that firms absorbing higher collection risk relative to their non-operating income buffer may face greater financial constraints or deteriorating credit quality among their customers. We calculate the signal using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both RECD and NOPIO variables, and exclude observations where the denominator equals zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CRA signal. Panel A plots the time-series of the mean, median, and interquartile range for CRA. On average, the cross-sectional mean (median) CRA is 0.13 (-0.01) over the 1971 to 2024 sample, where the starting date is determined by the availability of the input CRA data. The signal's interquartile range spans -0.53 to 0.33. Panel B of Figure 1 plots the time-series of the coverage of the CRA signal for the CRSP universe. On average, the CRA signal is available for 3.63% of CRSP names, which on average make up 4.61% of total market capitalization.

4 Does CRA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CRA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CRA portfolio and sells the low CRA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CRA strategy earns an average return of 0.21% per month with a t-statistic of 2.40. The annualized Sharpe ratio of the strategy is 0.33. The alphas range from 0.18% to 0.24% per month and have t-statistics exceeding 1.94 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.12, with a t-statistic of -2.81 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 352 stocks and an average market capitalization of at least $881 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 12 bps/month with a t-statistics of 2.56. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -12-25bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.03. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CRA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the CRA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CRA, as well as average returns and alphas for long/short trading CRA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the CRA strategy achieves an average return of 25 bps/month with a t-statistic of 2.28. Among these large cap stocks, the alphas for the CRA strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.14 and 2.69.

5 How does CRA perform relative to the zoo?

Figure 2 puts the performance of CRA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the CRA strategy falls in the distribution. The CRA strategy's gross (net) Sharpe ratio of 0.33 (0.26) is greater than 70% (84%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CRA strategy (red line).[2] Ignoring trading costs, a \$1 invested in the CRA strategy would have yielded \$2.38 which ranks the CRA strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CRA strategy would have yielded \$1.50 which ranks the CRA strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CRA relative to those. Panel A shows that the CRA strategy gross alphas fall between the 48 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197106 to 202406 sample. For example, 43%

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CRA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CRA ranks between the 65 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CRA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CRA with 208 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CRA or at least to weaken the power CRA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CRA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CRA signal in these Fama-MacBeth regressions exceed 0.11, with the minimum t-statistic occurring when controlling for Composite debt issuance. Controlling for all six closely related anomalies, the t-statistic on CRA is 0.74.

Similarly, Table 5 reports results from spanning tests that regress returns to the CRA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CRA strategy earns alphas that range from 19-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.98, which is achieved when controlling for Composite debt issuance. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CRA trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.16.

7 Does CRA add relative to the whole zoo?

Finally, we can ask how much adding CRA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CRA signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1083.64, while $1 investment in the combination strategy that includes CRA grows to $1474.35.

8 Conclusion

This study provides robust evidence that Collection Risk Absorption (CRA) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CRA-based trading strategies generate economically and statistically significant abnormal returns even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CRA delivers an annualized net Sharpe ratio of 0.26, which, while modest, remains econom-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CRA is available.

ically meaningful in the context of increasingly efficient markets. More importantly, the strategy's ability to generate a monthly alpha of 17 basis points (t-statistic = 1.89) relative to the Fama-French five-factor model plus momentum, and 23 basis points (t-statistic = 2.16) when controlling for six additional related factors, suggests that CRA captures a distinct dimension of risk or mispricing not fully explained by existing factors. The persistence of these abnormal returns after accounting for transaction costs and controlling for related strategies such as composite debt issuance, asset growth, and pension funding status underscores the signal's incremental information content.

From a practical perspective, these findings have important implications for both asset managers and corporate decision-makers. For portfolio managers, CRA offers a viable strategy for enhancing risk-adjusted returns, particularly when combined with other established factors in a multi-signal framework. For corporate managers and analysts, understanding how collection risk absorption activities influence market valuations can inform capital allocation decisions and risk management strategies.

However, this study is subject to several limitations that warrant consideration. First, while our results are robust to standard risk adjustments, the possibility remains that CRA proxies for an unidentified systematic risk factor rather than a market inefficiency. Second, the relatively modest Sharpe ratio and marginal statistical significance after transaction costs suggest that implementation challenges may limit the strategy's practical viability, particularly for large-scale institutional investors facing capacity constraints. Third, our analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains unexplored.

Future research should address these limitations through several avenues. First, investigating the economic mechanisms underlying the CRA effect would enhance our understanding of whether it represents risk compensation or behavioral mispricing. Second, examining the signal's performance across different market regimes, particularly during financial crises when collection risks may be heightened, could provide insights into its time-varying nature. Third, extending the analysis to international markets would test the robustness of the CRA effect across different institutional and regulatory environments. Finally, exploring potential interactions between CRA and other corporate financial policies, such as working capital management and credit extension strategies, could yield a more comprehensive understanding of how firms' collection risk management affects their market valuations. Such investigations would not only advance our theoretical understanding of asset pricing but also provide practical guidance for investment strategy development and corporate financial management.

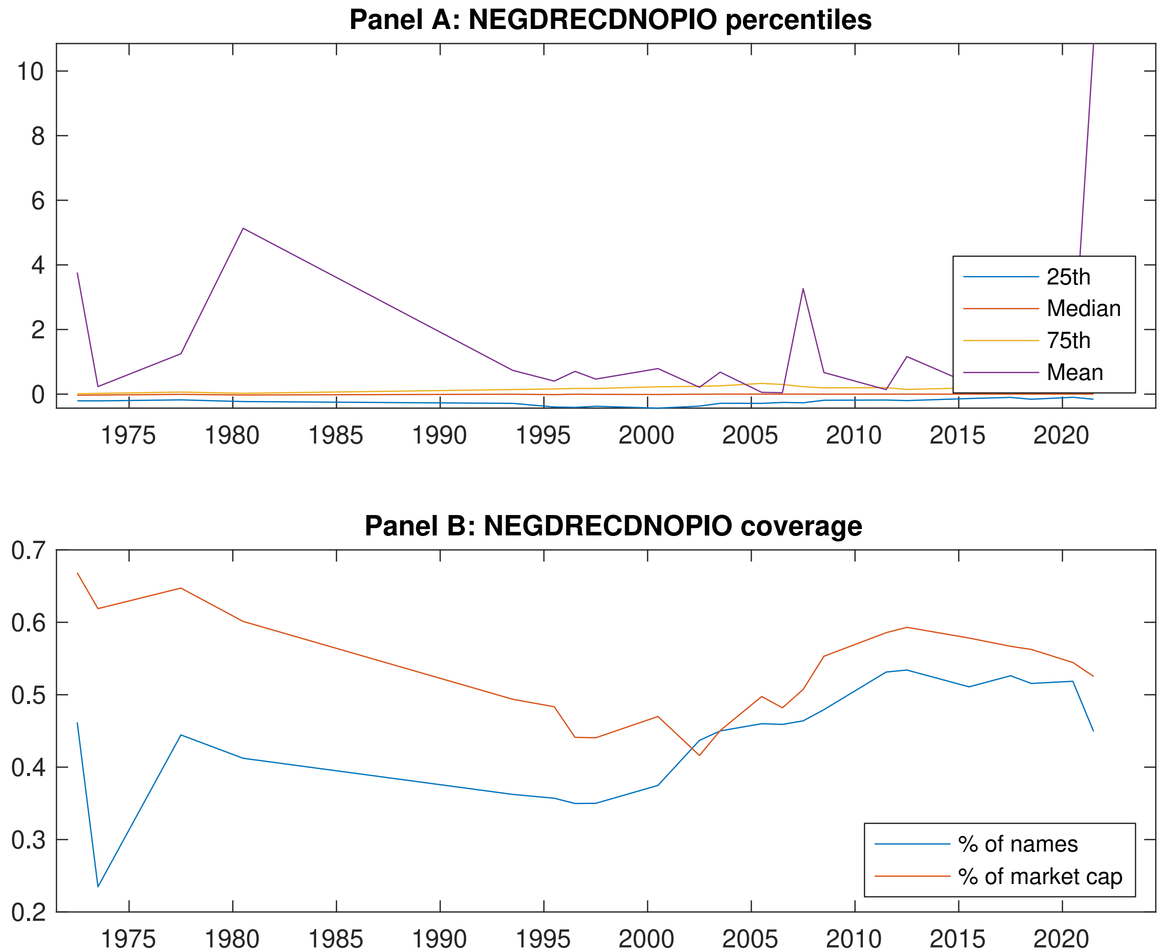


Figure 1: Times series of CRA percentiles and coverage.
This figure plots descriptive statistics for CRA. Panel A shows cross-sectional percentiles of CRA over the sample. Panel B plots the monthly coverage of CRA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CRA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Excess returns and alphas on CRA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.62	0.55	0.74	0.76	0.21
	[2.62]	[3.15]	[2.84]	[4.05]	[3.68]	[2.40]
α_{CAPM}	-0.13	-0.03	-0.07	0.15	0.09	0.23
	[-2.03]	[-0.52]	[-0.97]	[2.38]	[1.32]	[2.54]
α_{FF3}	-0.08	-0.02	-0.04	0.13	0.12	0.20
	[-1.29]	[-0.29]	[-0.56]	[2.20]	[1.75]	[2.24]
α_{FF4}	-0.08	0.03	0.00	0.15	0.09	0.18
	[-1.30]	[0.52]	[0.00]	[2.48]	[1.36]	[1.94]
α_{FF5}	-0.10	-0.02	0.04	0.07	0.14	0.24
	[-1.57]	[-0.32]	[0.62]	[1.21]	[1.95]	[2.59]
α_{FF6}	-0.10	0.02	0.07	0.10	0.11	0.22
	[-1.57]	[0.37]	[0.96]	[1.63]	[1.62]	[2.33]
Panel B: Fama and French (2018) 6-factor model loadings for CRA-sorted portfolios						
β_{MKT}	1.06	1.04	0.97	1.00	1.03	-0.03
	[69.67]	[75.90]	[56.52]	[69.59]	[62.71]	[-1.19]
β_{SMB}	0.13	-0.03	-0.07	-0.14	0.18	0.05
	[5.60]	[-1.30]	[-2.75]	[-6.47]	[7.30]	[1.62]
β_{HML}	-0.16	-0.07	-0.00	-0.07	-0.07	0.08
	[-5.43]	[-2.86]	[-0.04]	[-2.64]	[-2.35]	[2.03]
β_{RMW}	0.07	-0.00	-0.12	0.01	-0.05	-0.12
	[2.38]	[-0.17]	[-3.54]	[0.30]	[-1.49]	[-2.81]
β_{CMA}	-0.01	0.05	-0.19	0.28	-0.01	0.01
	[-0.33]	[1.35]	[-3.72]	[6.78]	[-0.12]	[0.14]
β_{UMD}	0.00	-0.06	-0.04	-0.04	0.03	0.03
	[0.09]	[-4.24]	[-2.16]	[-2.63]	[1.92]	[1.40]
Panel C: Average number of firms (n) and market capitalization (me)						
n	433	352	415	354	429	
me ($\$10^6$)	1059	1438	1497	1467	881	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CRA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197106 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.23	0.20	0.18	0.24	0.22
			[2.40]	[2.54]	[2.24]	[1.94]	[2.59]	[2.33]
Quintile	NYSE	EW	0.12	0.14	0.14	0.13	0.15	0.14
			[2.56]	[2.91]	[2.95]	[2.64]	[3.08]	[2.82]
Quintile	Name	VW	0.22	0.23	0.20	0.19	0.24	0.23
			[2.38]	[2.39]	[2.08]	[1.97]	[2.44]	[2.33]
Quintile	Cap	VW	0.18	0.21	0.16	0.14	0.16	0.15
			[2.06]	[2.37]	[1.90]	[1.62]	[1.84]	[1.62]
Decile	NYSE	VW	0.30	0.29	0.28	0.29	0.36	0.36
			[2.51]	[2.34]	[2.28]	[2.30]	[2.91]	[2.87]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.17	0.18	0.16	0.15	0.18	0.17
			[1.87]	[2.03]	[1.74]	[1.60]	[1.98]	[1.89]
Quintile	NYSE	EW	-0.12					
			[-2.03]					
Quintile	Name	VW	0.17	0.18	0.15	0.15	0.17	0.17
			[1.84]	[1.85]	[1.55]	[1.53]	[1.78]	[1.78]
Quintile	Cap	VW	0.14	0.17	0.13	0.11	0.12	0.12
			[1.57]	[1.90]	[1.45]	[1.30]	[1.37]	[1.31]
Decile	NYSE	VW	0.25	0.24	0.23	0.24	0.29	0.30
			[2.02]	[1.93]	[1.85]	[1.91]	[2.33]	[2.37]

Table 3: Conditional sort on size and CRA

This table presents results for conditional double sorts on size and CRA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CRA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CRA and short stocks with low CRA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197106 to 202406.

Panel A: portfolio average returns and time-series regression results

		CRA Quintiles					CRA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.71	0.76	0.66	0.97	0.86	0.15	0.17	0.17	0.14	0.14	0.12
		[2.53]	[2.89]	[2.43]	[3.64]	[3.14]	[1.73]	[1.99]	[1.95]	[1.63]	[1.59]	[1.37]
	(2)	0.76	0.82	0.77	0.80	0.83	0.07	0.08	0.09	0.13	0.14	0.16
		[2.90]	[3.23]	[3.05]	[3.18]	[3.24]	[0.74]	[0.94]	[1.06]	[1.44]	[1.49]	[1.77]
	(3)	0.77	0.81	0.66	0.87	0.77	0.00	0.02	0.04	0.04	0.11	0.11
		[3.07]	[3.44]	[2.73]	[3.72]	[3.15]	[0.04]	[0.16]	[0.36]	[0.41]	[1.05]	[1.06]
	(4)	0.68	0.79	0.74	0.75	0.74	0.06	0.00	0.01	-0.01	0.10	0.07
		[3.02]	[3.38]	[3.31]	[3.39]	[3.03]	[0.60]	[0.02]	[0.14]	[-0.07]	[1.00]	[0.76]
	(5)	0.46	0.58	0.51	0.76	0.72	0.25	0.30	0.26	0.25	0.25	0.25
		[2.25]	[2.97]	[2.57]	[4.08]	[3.72]	[2.28]	[2.69]	[2.31]	[2.23]	[2.18]	[2.14]

Panel B: Portfolio average number of firms and market capitalization

		CRA Quintiles					CRA Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	213	214	214	213	212	18	20	17	20	18
	(2)	64	64	64	64	64	36	37	36	37	36
	(3)	45	46	46	46	46	64	64	63	64	65
	(4)	39	38	38	39	39	142	147	143	145	140
	(5)	35	35	35	35	35	873	1115	1187	1100	754

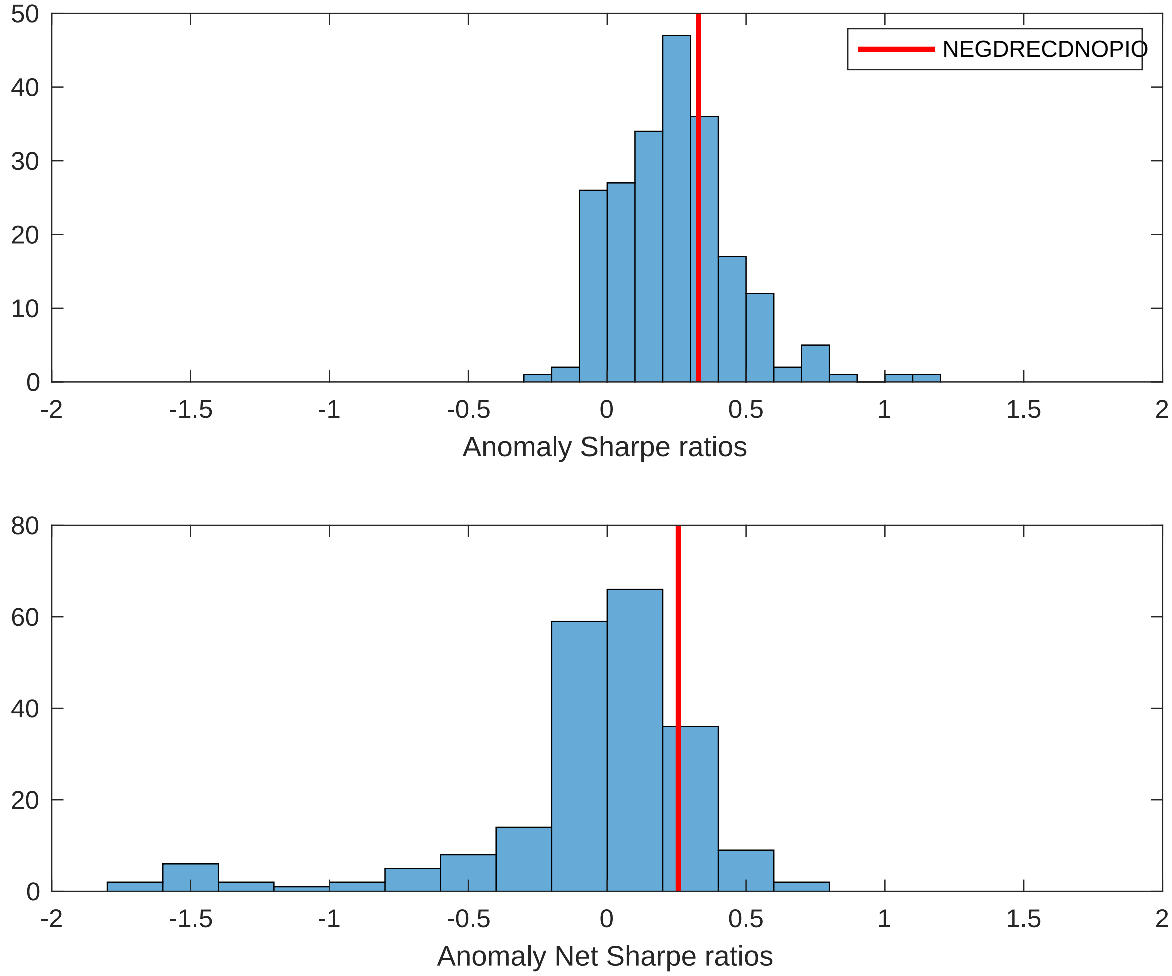


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CRA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

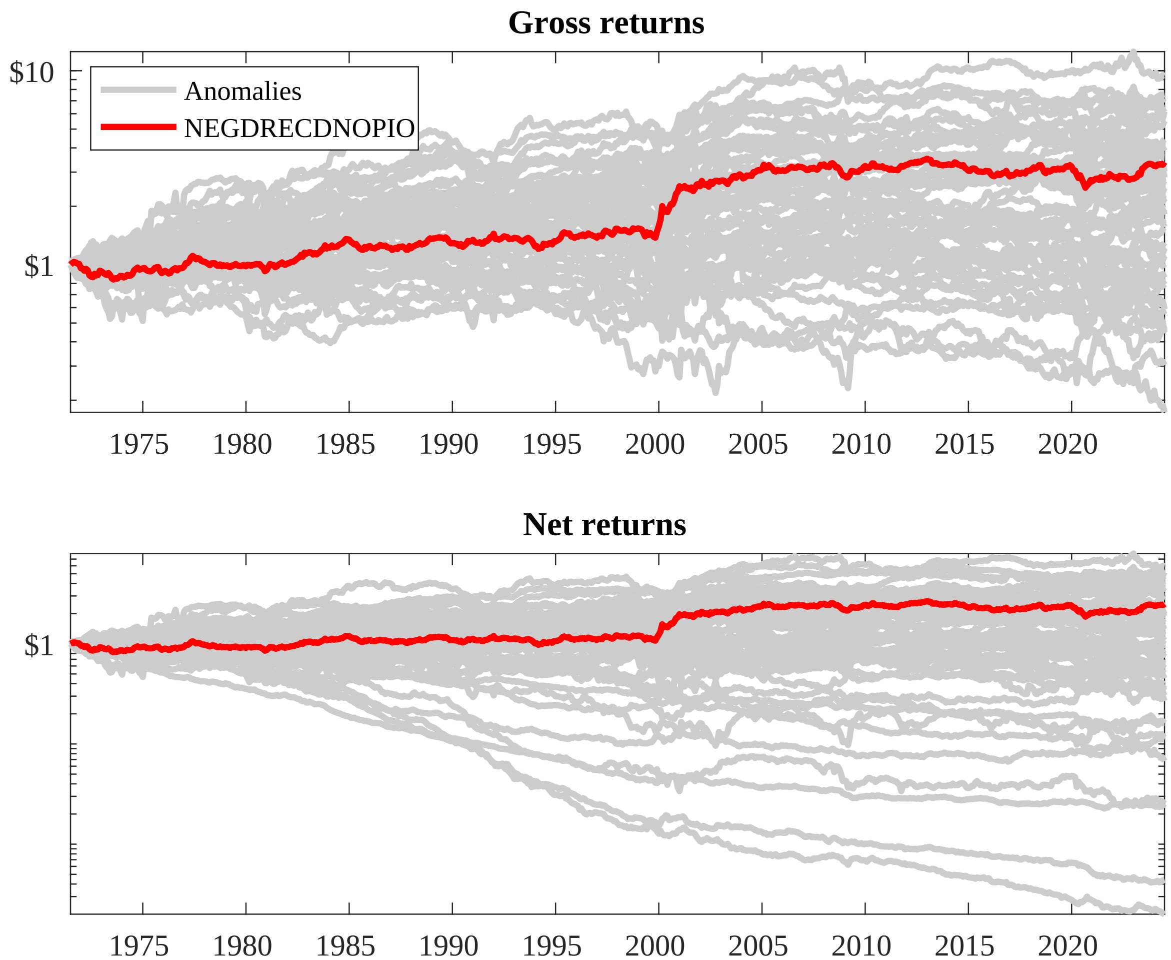
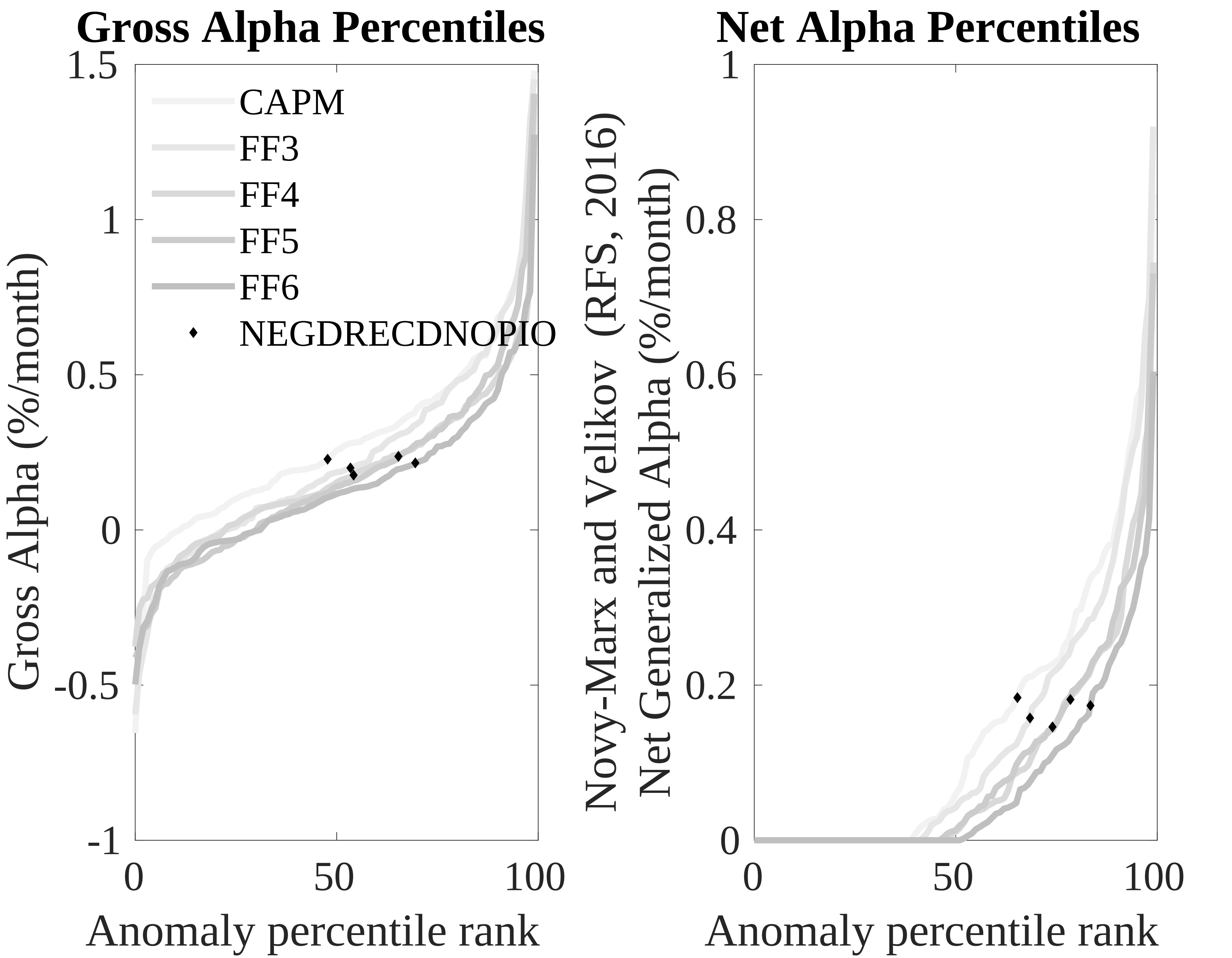


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CRA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDRECDNOPIO
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

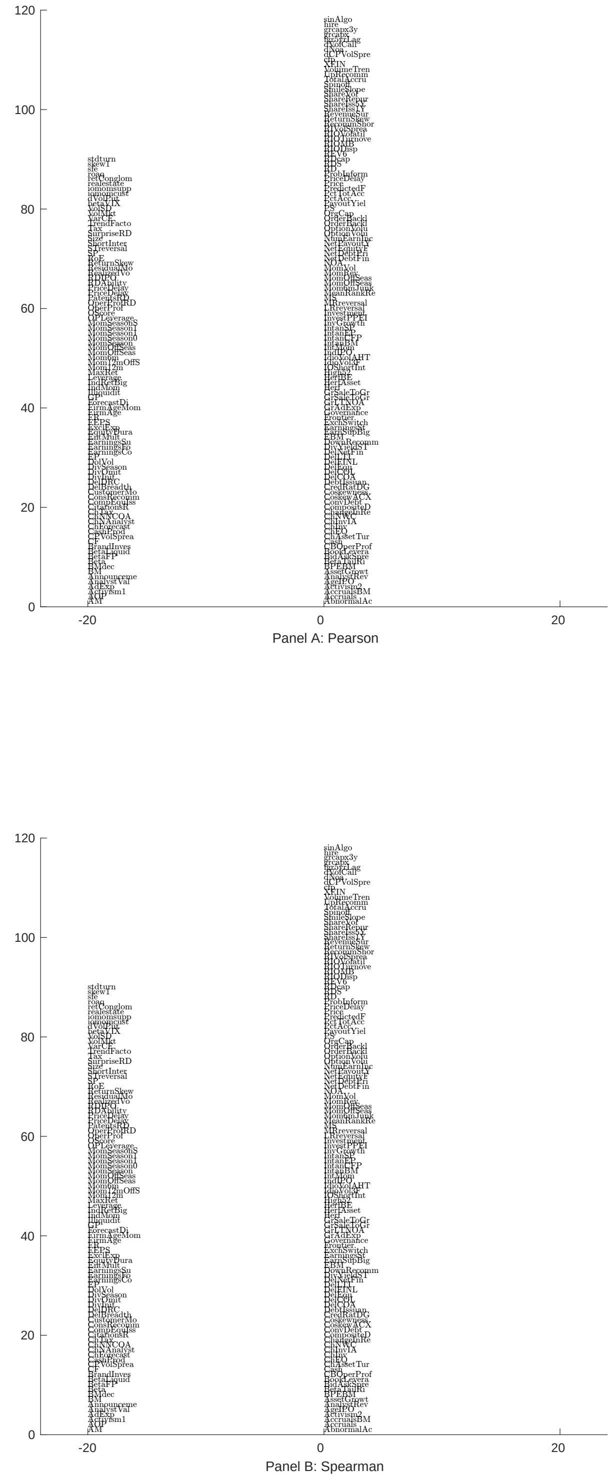


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 208 filtered anomaly signals with CRA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

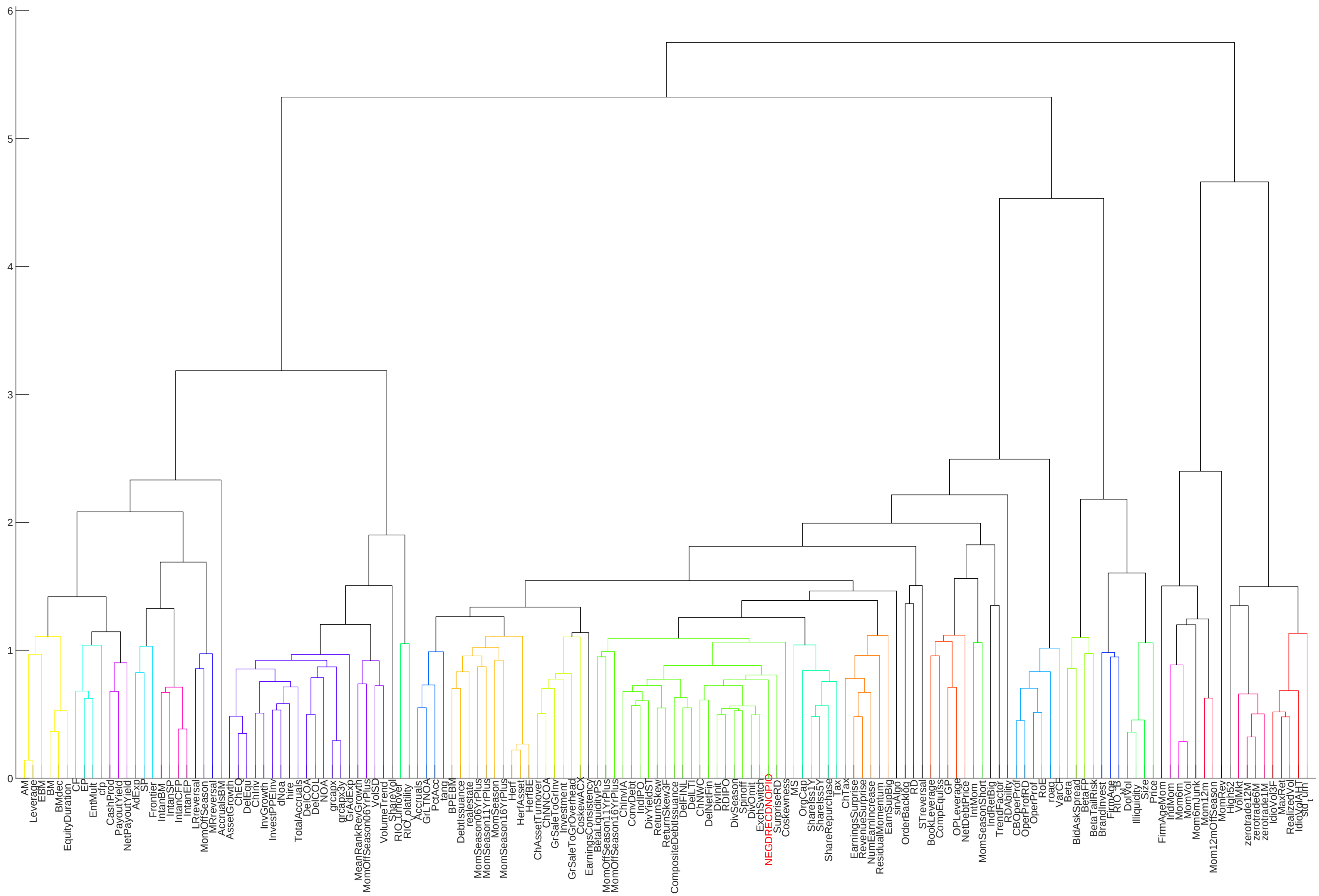


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

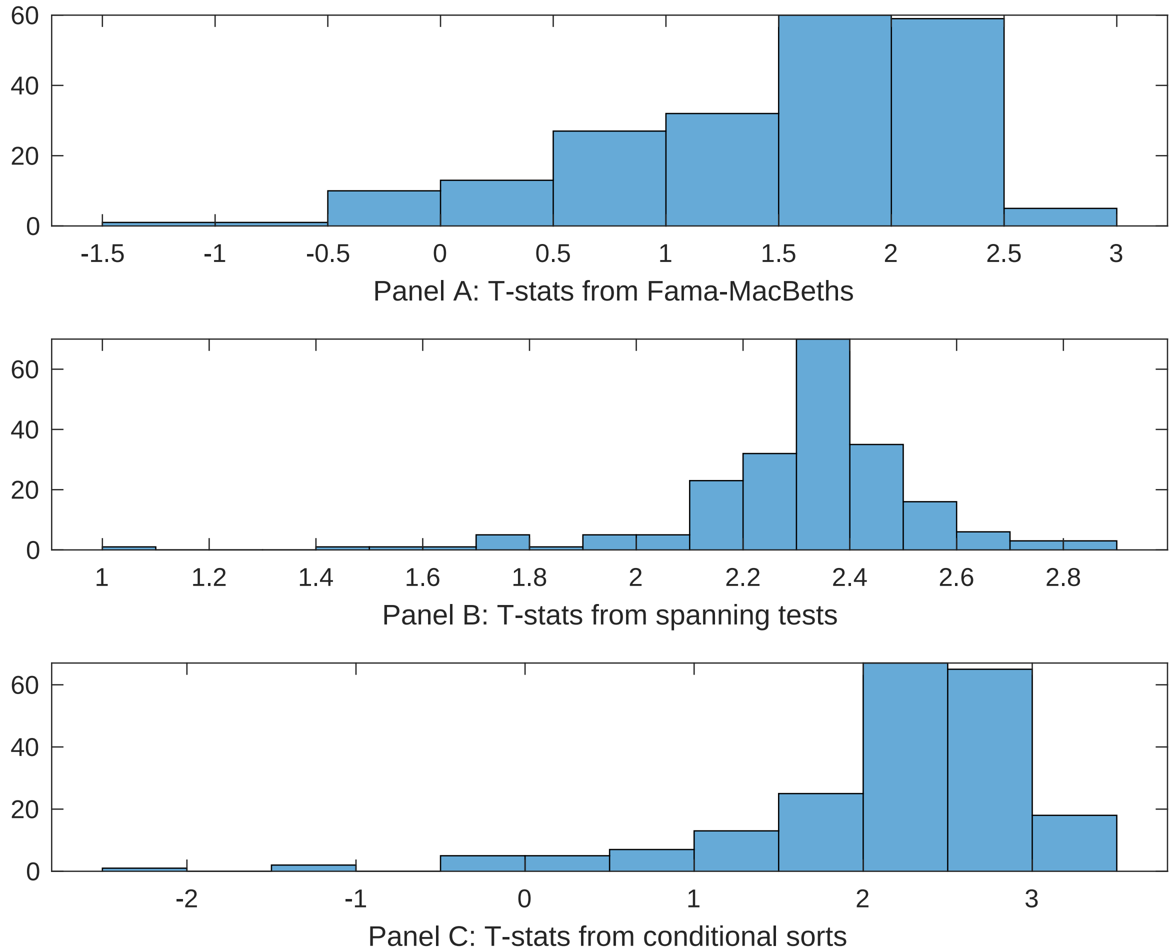


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CRA conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CRA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CRA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CRA. Stocks are finally grouped into five CRA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CRA trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CRA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CRA} CRA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.13	0.18	0.13	0.14	0.12	-0.99	-0.98
	[5.61]	[6.70]	[5.42]	[5.51]	[5.05]	[-9.48]	[-7.17]
CRA	0.67	0.90	0.69	0.74	0.12	0.39	0.59
	[0.11]	[1.84]	[1.41]	[1.49]	[1.50]	[0.71]	[0.74]
Anomaly 1	0.11						0.32
	[5.76]						[1.40]
Anomaly 2		0.51					0.29
		[4.61]					[0.28]
Anomaly 3			0.15				0.47
			[9.65]				[1.50]
Anomaly 4				0.10			-0.22
				[8.16]			[-0.92]
Anomaly 5					0.60		0.51
					[1.06]		[0.90]
Anomaly 6						0.47	0.49
						[8.01]	[5.99]
# months	636	636	636	636	516	631	516
$\bar{R}^2(\%)$	0	0	0	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CRA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CRA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Composite debt issuance, Growth in book equity, change in net operating assets, Asset growth, Pension Funding Status, Trend Factor. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197106 to 202406.

Intercept	0.22	0.21	0.21	0.22	0.28	0.19	0.23
	[2.38]	[2.28]	[2.20]	[2.33]	[2.59]	[1.98]	[2.16]
Anomaly 1	10.47						14.54
	[2.40]						[2.92]
Anomaly 2		14.40					22.14
		[2.86]					[3.20]
Anomaly 3			8.55				3.71
			[1.58]				[0.57]
Anomaly 4				4.39			-13.51
				[0.80]			[-1.82]
Anomaly 5					-11.12		-7.44
					[-2.87]		[-1.91]
Anomaly 6						3.34	5.55
						[1.58]	[2.39]
mkt	-1.88	-2.05	-2.46	-2.50	-2.84	-2.81	-0.96
	[-0.86]	[-0.95]	[-1.13]	[-1.15]	[-1.11]	[-1.29]	[-0.37]
smb	3.99	5.26	5.80	5.37	4.24	5.60	2.22
	[1.20]	[1.62]	[1.78]	[1.63]	[1.07]	[1.70]	[0.56]
hml	7.38	7.12	8.11	8.66	8.80	8.97	5.23
	[1.76]	[1.70]	[1.94]	[2.08]	[1.88]	[2.15]	[1.11]
rmw	-12.82	-11.27	-11.72	-11.85	-9.83	-10.58	-8.63
	[-3.02]	[-2.67]	[-2.76]	[-2.79]	[-2.03]	[-2.45]	[-1.77]
cma	-2.21	-13.47	-5.84	-4.86	-6.66	0.26	-18.64
	[-0.35]	[-1.69]	[-0.78]	[-0.53]	[-0.95]	[0.04]	[-1.73]
umd	2.66	2.73	2.81	3.18	5.20	2.45	2.36
	[1.23]	[1.27]	[1.30]	[1.47]	[2.11]	[1.11]	[0.94]
# months	636	636	636	636	517	632	517
$\bar{R}^2(\%)$	4	4	3	3	3	3	6

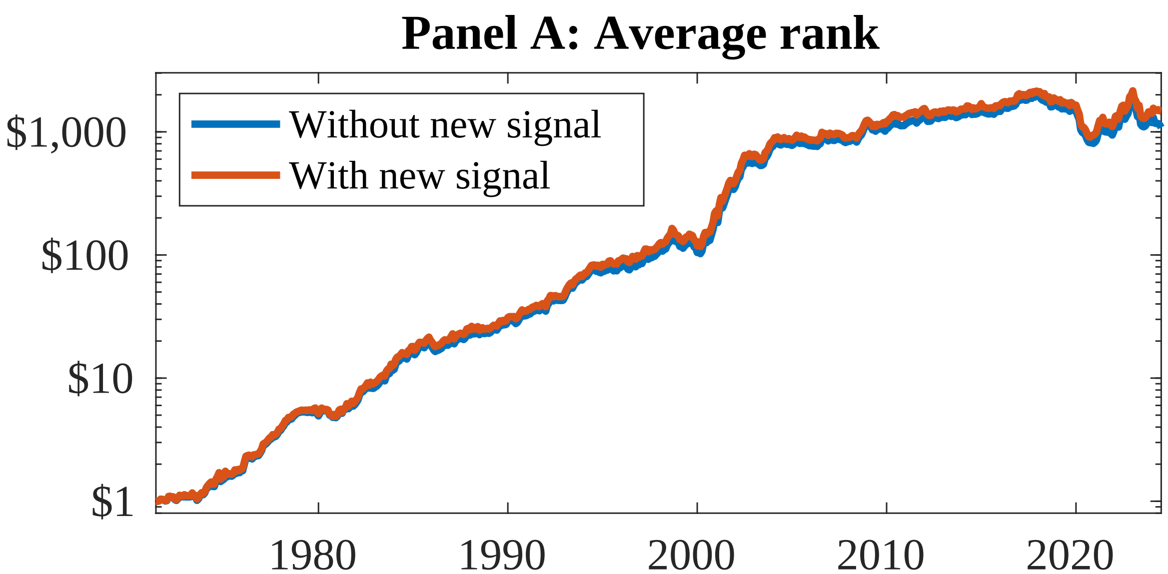


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CRA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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